



**Tertiary Infotech Academy Pte Ltd**

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# LESSON PLAN

For

## **Codex Masterclass**

Course Code: C989

Conducted by

**Tertiary Infotech Academy Pte Ltd**

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**Version 1.0**

## DOCUMENT VERSION CONTROL RECORD

| Version Number | Effective Date of Release | Summary of Included Changes                                                                                                 | Author         |
|----------------|---------------------------|-----------------------------------------------------------------------------------------------------------------------------|----------------|
| 1.0            | 26 Sep 2026               | First release of the one-day Codex Masterclass: Topics 1–3 and Labs 2–11 of the Cook & Bake Academy build, 9:30am – 5:30pm. | Dr. Alfred Ang |

## TABLE OF CONTENTS

|                                                                                  |          |
|----------------------------------------------------------------------------------|----------|
| <b>Course Overview.....</b>                                                      | <b>4</b> |
| <b>Learning Outcomes.....</b>                                                    | <b>4</b> |
| <b>Daily Schedule.....</b>                                                       | <b>4</b> |
| Day 1 — Build, Assist and Govern.....                                            | 4        |
| <b>Topic-by-topic Breakdown.....</b>                                             | <b>6</b> |
| Topic 1 — Fundamentals: Chat, Work and Codex · Slides 11–43.....                 | 6        |
| Lab 2 — Plan and Build the Site with /plan · Slides 26–29.....                   | 6        |
| Lab 3 — Project Rules and a Sign-up Form for Every Course · Slides 33–36.....    | 6        |
| Lab 4 — Publish the Site: GitHub Pages and Sites · Slides 38–42.....             | 6        |
| Topic 2 — Tools and the SQLite RAG Assistant · Slides 44–73.....                 | 7        |
| Lab 5 — Turn the Brochures into a SQLite Knowledge Base · Slides 51–54.....      | 7        |
| Lab 6 — Build the Course Assistant and Drive It with /goal · Slides 56–60.....   | 7        |
| Lab 7 — Add ChatGPT Mode, Then Try to Break It · Slides 64–67.....               | 7        |
| Lab 8 — QA the Whole Site with @Computer Use · Slides 69–72.....                 | 7        |
| Topic 3 — Skills and Hooks · Slides 74–102.....                                  | 8        |
| Lab 9 — Install Community Skills from skills.sh · Slides 80–85.....              | 8        |
| Lab 10 — Create Custom Codex Skills · Slides 88–93.....                          | 8        |
| Lab 11 — A Workshop Popup and a Hook That Checks Every Edit · Slides 98–102..... | 8        |
| <b>Resources Required.....</b>                                                   | <b>8</b> |

## Course Overview

A one-day, hands-on course in which learners use Codex, the agentic coding surface in the ChatGPT desktop app, to take one business — Cook & Bake Academy Singapore, a cooking and baking school — from a written brief to a live, assisted and governed website. Codex plans and builds the site and a sign-up form for every course, publishes it, turns the brochures into a SQLite knowledge base behind a RAG course assistant, red-teams and QA-tests it with Computer Use, then adds community and custom skills, a workshop popup and a hook that checks every edit. Every lab starts from what the previous lab produced.

Course code C989 · 1 day · 7.5 instructional hours (9:30am – 5:30pm) · 106-slide deck v1.0 · 10 labs (Labs 2–11), each in its own folder.

## Learning Outcomes

By the end of this course, participants will be able to:

- **LO1.** Analyse agentic AI applications — ChatGPT, ChatGPT Work and Codex — and their strengths, limitations and suitability for a business problem.
- **LO2.** Correlate the design of an agent's instructions, tools and retrieval with the efficiency and quality of the result.
- **LO3.** Assess the effectiveness, safety and reliability of agent-built work using golden-question evaluation, hooks, Computer Use testing and human review.
- **LO4.** Evaluate comparative effectiveness and recommend a governed way to run agentic AI applications in a real business, using observable evidence.

## Daily Schedule

The day runs 9:30am – 5:30pm: 7.5 instructional hours with a single 30-minute lunch. The Slides column gives the deck pages for each teaching block.

### Day 1 — Build, Assist and Govern · 9:30am – 5:30pm

| Time              | Duration | Topic / Activity                                                                                                                                               | Method                    | Slides |
|-------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|--------|
| 9:30 AM–9:50 AM   | 20 min   | Welcome, trainer, ground rules, the Cook & Bake scenario, the right surface for each job, course outline, lab materials and schedule                           | Presentation , discussion | 1–10   |
| 9:50 AM–10:15 AM  | 25 min   | Topic 1 — evolution of AI engineering, harness engineering and the agent loop, Chat, Work and Codex, GPT-6 models, desktop app, plugins, Work research, @Sites | Presentation , live demo  | 11–21  |
| 10:15 AM–10:25 AM | 10 min   | Codex: projects, folders and permissions, the 7-step workflow, /plan                                                                                           | Presentation , live demo  | 22–25  |
| 10:25 AM–11:05    | 40 min   | Lab 2 — Plan and build the site with                                                                                                                           | Hands-on                  | 26–29  |

|                   |        |                                                                             |                          |       |
|-------------------|--------|-----------------------------------------------------------------------------|--------------------------|-------|
| AM                |        | /plan                                                                       | lab                      |       |
| 11:05 AM–11:10 AM | 5 min  | Context engineering with AGENTS.md; a sign-up form with no backend          | Presentation             | 30–32 |
| 11:10 AM–11:40 AM | 30 min | Lab 3 — Project rules and a sign-up form for every course                   | Hands-on lab             | 33–36 |
| 11:40 AM–11:45 AM | 5 min  | Publishing: GitHub Pages or Sites                                           | Presentation             | 37    |
| 11:45 AM–12:10 PM | 25 min | Lab 4 — Publish the site: GitHub Pages and Sites                            | Hands-on lab             | 38–42 |
| 12:10 PM–12:15 PM | 5 min  | Topic 1 recap                                                               | Discussion               | 43    |
| 12:15 PM–12:45 PM | 30 min | Lunch                                                                       | —                        | —     |
| 12:45 PM–1:00 PM  | 15 min | Topic 2 — slash commands, /goal, RAG, SQLite in the browser, architecture   | Presentation , live demo | 44–50 |
| 1:00 PM–1:25 PM   | 25 min | Lab 5 — Turn the brochures into a SQLite knowledge base                     | Hands-on lab             | 51–54 |
| 1:25 PM–1:30 PM   | 5 min  | Measuring before trusting: the golden-question set                          | Presentation             | 55    |
| 1:30 PM–2:05 PM   | 35 min | Lab 6 — Build the course assistant and drive it with /goal                  | Hands-on lab             | 56–60 |
| 2:05 PM–2:10 PM   | 5 min  | Search mode vs ChatGPT mode; keys and static sites                          | Presentation             | 61–63 |
| 2:10 PM–2:35 PM   | 25 min | Lab 7 — Add ChatGPT mode, then try to break it                              | Hands-on lab             | 64–67 |
| 2:35 PM–2:40 PM   | 5 min  | The Computer Use tool                                                       | Presentation             | 68    |
| 2:40 PM–3:05 PM   | 25 min | Lab 8 — QA the whole site with @Computer Use                                | Hands-on lab             | 69–72 |
| 3:05 PM–3:10 PM   | 5 min  | Topic 2 recap                                                               | Discussion               | 73    |
| 3:10 PM–3:20 PM   | 10 min | Topic 3 — agent skills, where skills live, SKILL.md vs AGENTS.md, skills.sh | Presentation , live demo | 74–79 |
| 3:20 PM–3:45 PM   | 25 min | Lab 9 — Install community skills from skills.sh                             | Hands-on lab             | 80–85 |
| 3:45 PM–3:50 PM   | 5 min  | Creating your own skill; anatomy of kb-update                               | Presentation             | 86–87 |
| 3:50 PM–4:15 PM   | 25 min | Lab 10 — Create custom Codex skills                                         | Hands-on lab             | 88–93 |

|                 |        |                                                                                 |              |         |
|-----------------|--------|---------------------------------------------------------------------------------|--------------|---------|
| 4:15 PM–4:25 PM | 10 min | Hooks: triggered by events — timer vs hook vs schedule                          | Presentation | 94–97   |
| 4:25 PM–5:00 PM | 35 min | Lab 11 — A workshop popup and a hook that checks every edit                     | Hands-on lab | 98–102  |
| 5:00 PM–5:30 PM | 30 min | Course summary: how each feature is triggered, wrap-up, Q&A and course feedback | Discussion   | 103–105 |

## Topic-by-topic Breakdown

### Topic 1 — Fundamentals: Chat, Work and Codex • Slides 11–43

- How AI engineering got here, and why the harness matters
- OpenAI's products, the GPT-6 models and the desktop app
- Projects, permissions, /plan and the 7-step workflow
- Build the site with Codex, add sign-ups, publish it

### Lab 2 — Plan and Build the Site with /plan • Slides 26–29

- **Surface:** Codex
- **Time:** 40 minutes
- **Features practised:** Add a folder · GPT-6 Sol · /plan · the 7-step workflow
- **Output:** A running site: hero with photos, 20 course cards, filters, search, campuses
- **Evidence:** Plan mode showed numbered steps AND questions before any edit. No horizontal scroll at 375px wide.
- **Lab folder:** labs/lab-02-plan-and-build-the-site/

### Lab 3 — Project Rules and a Sign-up Form for Every Course • Slides 33–36

- **Surface:** Codex
- **Time:** 30 minutes
- **Features practised:** /init · AGENTS.md · static-site forms · consent
- **Output:** AGENTS.md, a validated sign-up form on every card, and a staff CSV export
- **Evidence:** AGENTS.md has the four sections and is under 60 lines. admin.html exports a CSV with the 12 columns in the spec.
- **Lab folder:** labs/lab-03-rules-and-signup-forms/

### Lab 4 — Publish the Site: GitHub Pages and Sites • Slides 38–42

- **Surface:** Codex → GitHub Pages → @Sites
- **Time:** 25 minutes
- **Features practised:** git · GitHub Pages (deploy from a branch) · @Sites access control
- **Output:** A public GitHub Pages site for customers and a private Sites copy for investors
- **Evidence:** Codex listed the files and none is a secret or an export. You can say which URL is for customers and which for investors.

- **Lab folder:** labs/lab-04-publish-the-site/

## Topic 2 — Tools and the SQLite RAG Assistant • Slides 44–73

- Codex commands, /goal and tools
- RAG: answering from your own documents
- A SQLite knowledge base that runs in the browser
- ChatGPT mode, red-teaming and Computer Use QA

### Lab 5 — Turn the Brochures into a SQLite Knowledge Base • Slides 51–54

- **Surface:** Codex
- **Time:** 25 minutes
- **Features practised:** SQLite FTS5 · official SQLite WASM · nested AGENTS.md
- **Output:** data/academy.db — 140 searchable chunks and a courses table
- **Evidence:** data/academy.db exists (about 150 KB). Running the build twice gives the same counts.
- **Lab folder:** labs/lab-05-sqlite-knowledge-base/

### Lab 6 — Build the Course Assistant and Drive It with /goal • Slides 56–60

- **Surface:** Codex
- **Time:** 35 minutes
- **Features practised:** In-browser RAG · golden-question evaluation · /goal
- **Output:** A working course assistant that passes 30/30 golden questions
- **Evidence:** The assistant runs in the browser with no server code. "What is the cheapest course?" answers CUL-210, S\$160.
- **Lab folder:** labs/lab-06-course-assistant-and-goal/

### Lab 7 — Add ChatGPT Mode, Then Try to Break It • Slides 64–67

- **Surface:** Codex
- **Time:** 25 minutes
- **Features practised:** OpenAI Responses API · gpt-6-luna · grounded prompt · prompt injection
- **Output:** A two-mode assistant that stays grounded under attack
- **Evidence:** git grep "sk-" finds no key anywhere in the repo. A bad key falls back to search mode with a message.
- **Lab folder:** labs/lab-07-chatgpt-mode-and-red-team/

### Lab 8 — QA the Whole Site with @Computer Use • Slides 69–72

- **Surface:** Codex + Computer Use
- **Time:** 25 minutes
- **Features practised:** @Computer Use · evidence-based QA
- **Output:** A ranked defect table with screenshots, and one verified fix
- **Evidence:** Computer Use drove a real browser through the whole script. The top defect is fixed and the retest confirms it.
- **Lab folder:** labs/lab-08-qa-with-computer-use/

### Topic 3 — Skills and Hooks • Slides 74–102

- Skills: packaged expertise, from skills.sh or your own
- Custom Codex skills with \$skill-creator
- A workshop popup that turns visitors into sign-ups
- A hook that re-checks the site after every edit

#### Lab 9 — Install Community Skills from skills.sh • Slides 80–85

- **Surface:** Codex
- **Time: 25 minutes**
- **Features practised:** npx skills add · frontend-design · cybersecurity-analyst · anthropic-cybersecurity-skills
- **Output:** A polished UI, a ranked security review, and the skills lock file
- **Evidence:** Three skills are in .agents/skills/ and in the lock file. You fixed the top finding (e.g. a Content-Security-Policy).
- **Lab folder:** labs/lab-09-skills-from-skills-sh/

#### Lab 10 — Create Custom Codex Skills • Slides 88–93

- **Surface:** Codex
- **Time: 25 minutes**
- **Features practised:** \$skill-installer · \$skill-creator · implicit triggering
- **Output:** \$kb-update and \$course-brochure skills, and a new course live
- **Evidence:** project-setup is installed and listed, but was not run. \$kb-update fired on its own in a new chat.
- **Lab folder:** labs/lab-10-custom-codex-skills/

#### Lab 11 — A Workshop Popup and a Hook That Checks Every Edit • Slides 98–102

- **Surface:** Codex
- **Time: 35 minutes**
- **Features practised:** timed popup · 3-field form · hook on an event · trust
- **Output:** A 10-second workshop invite with a name, mobile and email form, and a trusted hook that runs npm run check after every edit
- **Evidence:** The invite opens after about 10 seconds, not before. npm run check passes every golden question, and the live site shows the invite.
- **Lab folder:** labs/lab-11-workshop-popup-and-a-hook/

## Resources Required

- Trainer: the v1.0 slide deck, this Lesson Plan, the Learner Guide, the lab pack (labs/, 10 folders: Labs 2–11) and the reference app.
- Each learner: a laptop with the ChatGPT desktop app on a paid plan that includes Codex and Sites; Node.js 22; Python 3; Git; and a GitHub account.
- A spend-limited OpenAI API key issued by the trainer for Lab 7, revoked after class.
- Internet access to chatgpt.com, api.openai.com, github.com, skills.sh, npmjs.com and images.unsplash.com.

- All fixtures are synthetic. Every learner email address resolves to the learner's own Gmail via plus-addressing.